# Real-Time Driver Drowsiness Detection Using OpenCV and MediaPipe

Caraga State University – College of Computing and Information Sciences
Group Members: Geuel Morillon, Eddie Rufo II Gause, John Pintasan

## 1. Title and Objectives of the Project

**Project title:** Real-Time Driver Drowsiness Detection Using OpenCV and MediaPipe.

**General objective:** To design and implement a working computer vision application that detects signs of drowsiness from a live webcam feed using classical OpenCV techniques.

**Specific objectives:**

- Detect the user's face in real time using a Haar cascade classifier.
- Locate the eye landmarks and compute the Eye Aspect Ratio (EAR) to measure how open the eyes are.
- Show a "DROWSY! WAKE UP!" alert when the eyes stay closed for 1.5 seconds, ignoring normal blinks.
- Display processed frames with boxes, landmarks and labels, and log each drowsy event to a CSV file.
- Test the accuracy, behavior and limitations of the system.

## 2. Description of the Problem or Use-Case

Drowsiness slows a person's reactions and attention. For a driver, nodding off even for a moment can lead to accidents. The same problem affects students and workers who stay up late and need to stay alert.

This project is a low-cost proof of concept for a drowsiness monitor that needs only a laptop and a webcam. It watches the user's eyes and warns them when the eyes stay closed longer than a normal blink.

## 3. Techniques Used

The Python application processes a live webcam feed and uses three laboratory topics on every frame:

**Webcam frame → (a) Preprocessing → (c) Haar face detection → crop → (f) Landmarks and EAR → Status**

**a. Image Operations.** Each frame is mirrored, converted to grayscale, smoothed with a 3×3 Gaussian blur to reduce noise, and enhanced with CLAHE (contrast-limited adaptive histogram equalization) to even out lighting. The detected face region is cropped and resized to 256×256 pixels before landmarking. OpenCV drawing functions add the boxes, points, text, red alert border and a small preprocessed-image preview.

**c. Haar Cascade Classifier.** OpenCV's pre-trained haarcascade_frontalface_default.xml, run with detectMultiScale (scaleFactor 1.2, minNeighbors 5, minimum size 100×100 px), finds faces in the preprocessed frame. The largest face is used and drawn as a blue bounding box. If the detector misses for a few frames, the last box is kept briefly to avoid flickering. The box is padded into a square and passed to the landmark step.

**f. Facial Landmark Detection.** MediaPipe's Face Landmarker, a pre-trained model used as provided, returns facial landmarks for the cropped face. Six points around each eye give the Eye Aspect Ratio:

$$EAR = ( |p2 - p6| + |p3 - p5| ) / ( 2 \times |p1 - p4| )$$

The EAR of both eyes is averaged. An open eye gives a higher value and a closed eye gives a value near zero. In a calibration test on one team member, EAR was about 0.31 with eyes open and 0.02 with eyes closed, so the threshold was set to 0.20. The decision rule is:

| Condition | Status shown | Display |
|---|---|---|
| EAR ≥ 0.20 | Awake | Green text |
| EAR < 0.20 for less than 1.5 s | Eyes closed | Orange text |
| EAR < 0.20 for 1.5 s or more | DROWSY! WAKE UP! | Red text, red border, entry in CSV log |
| No face found | NO FACE | Gray text |

## 4. Results, Accuracy, Behavior, and Limitations

The tests below were run with the final application on a laptop webcam. Each test shows what was done and what the program did.

### 4.1 Accuracy Test

| No. | Test | Result |
|---|---|---|
| 1 | Eyes open for 30 seconds | The program showed a consistent "Awake" status and was fully functional. |
| 2 | 20 normal blinks | The status stayed at "Awake" and "Eyes closed". No "DROWSY" alert was triggered (0 false alerts in 20 blinks). |
| 3 | Eyes closed for 2 seconds (10 trials) | The "DROWSY" alert appeared in every trial (10 out of 10). |
| 4 | Eyes closed for 1 second (10 trials) | The program showed "Awake" and "Eyes closed" only. No "DROWSY" alert was triggered in any trial (0 false alerts in 10), as expected since 1 second is below the 1.5-second limit. |
| 5 | Eyes closed, timed with a stopwatch (5 trials) | "Eyes closed" appeared as soon as the eyes closed, and the "DROWSY" alert followed after about 1.50 seconds, matching the setting. |

In all of these trials the system behaved as intended: every closure of 2 seconds was detected, and neither blinks nor 1-second closures caused a false alert.

### 4.2 Behavior Test

| No. | Test | Result |
|---|---|---|
| 1 | Opening the eyes after a drowsy alert | The program returned to the "Awake" status as soon as the eyes were opened. |
| 2 | Alert records | Each drowsy alert was recorded in drowsiness_log.csv (timestamp, EAR and closed duration), including when the alert was held for about 5 seconds. |
| 3 | Face covered | "NO FACE" appeared for a few frames, and the program went back to detecting "Awake" or "Eyes closed" once the face was visible again. |
| 4 | Two faces in view | The program tracked only the face closer to the camera (Figures 1 and 2). |




**Figure 1.** *Two faces in view: only the face closer to the camera (right) is tracked.*




**Figure 2.** *Two faces in view: the closer face (center) is tracked, not the other.*

### 4.3 Limitations

These tests show the conditions under which the system stops working. In every failing case below the screen shows "NO FACE", meaning no face was detected.

**Lighting.** A face is detected only if the lighting is enough to show the eyes, nose and mouth.

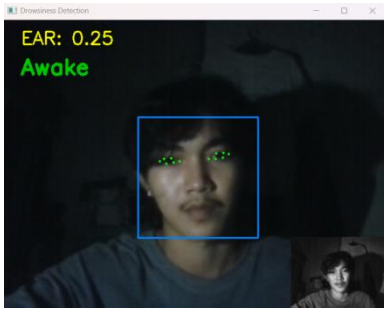



**Figure 3.** *Dim light: face detected (EAR 0.25, Awake).*

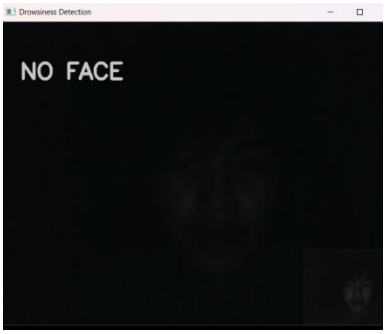



**Figure 4.** *Very dark: no face detected.*




**Figure 5.** *Bright light: face detected (EAR 0.28, Awake).*

**Distance.** A face is detected from around 1 foot to 4 feet from the camera.




**Figure 6.** *Too close: no face detected.*




**Figure 7.** *Too far: no face detected.*




**Figure 8.** *Within range: face detected (EAR 0.24, Awake).*

**Face angle.** A face is not detected when it is tilted to about 30°, when looking down (for example at a phone), or when facing sideways.




**Figure 9.** *Head tilted about 30°: no face detected.*




**Figure 10.** *Looking down at a phone: no face detected.*

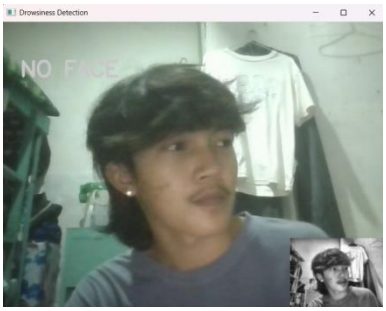



**Figure 11.** *Facing sideways: no face detected.*

**Eyewear and obstruction.** A face is not detected when glasses are worn, or when part of the face is covered.




**Figure 12.** *Wearing glasses: no face detected.*




**Figure 13.** *Hand covering one eye: no face detected.*




**Figure 14.** *Hand covering the mouth: no face detected.*

### 4.4 Discussion

**Accuracy.** When a face is detected, the EAR and timing logic worked reliably: all ten 2-second closures raised the alert, and none of the 20 blinks or ten 1-second closures did. These results come from a small number of trials, mostly on one team member in one room, so they show that the logic works but do not prove the same accuracy for other people or settings. In the screenshots the open-eye EAR was between 0.24 and 0.28, which is closer to the 0.20 threshold than the 0.31 measured in the earlier calibration, so the margin is smaller than expected.

**Behavior.** The program recovered immediately when the eyes opened or the face came back into view, recorded each alert in the CSV file, and followed only the closest face, which avoids mixing up several people.

**Limitations.** Every failure case (darkness, being too close or too far, head tilt, looking down, facing sideways, glasses and covered faces) ended in "NO FACE". Because the application runs the landmark step only inside the box found by the Haar cascade, the likely cause is that the Haar face detector is the weakest stage: it works best on a bright, unobstructed, mostly frontal face of reasonable size, and the code ignores faces smaller than 100×100 pixels. This is an interpretation of the results, not something measured separately. The EAR threshold was also calibrated on one person and may need adjusting for others.

## 5. Conclusion

The project met its goal of building a working real-time drowsiness detector with Python and OpenCV. It combines three laboratory topics: image operations for preprocessing and display, a Haar cascade for face detection, and facial landmark detection for measuring eye openness. It correctly told apart normal blinks and short closures from closures of 1.5 seconds or more, showed clear on-screen results, and logged each alert.

**Key learnings.** A simple measure such as the EAR, combined with a time limit, is enough to separate blinking from drowsiness. Preprocessing and lighting strongly affect how well classical detectors work, and testing under different conditions is needed to find where a system fails.

**Possible improvements:**

- Add a fallback that runs the landmark detector on the full frame when the Haar cascade finds no face, or replace the Haar step with a more robust face detector.
- Calibrate the EAR threshold automatically for each user, and test with more people and conditions.

- Add a head-pose check to tell looking down from closing the eyes, and add an audio alarm.
- Improve low-light performance, for example with a stronger enhancement step or an infrared camera.